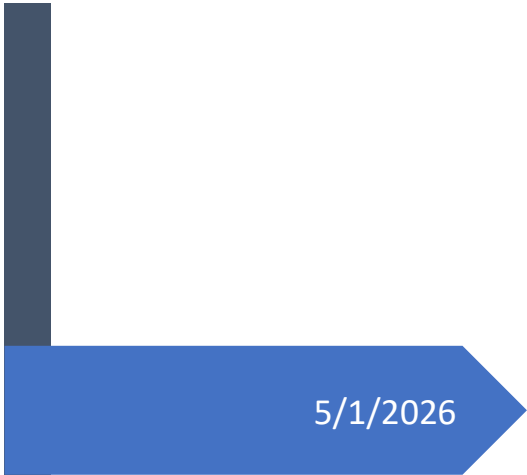


A dark blue vertical bar runs down the left side of the page. A blue arrow points to the right from this bar, containing the date.

5/1/2026

CYBERSHIELD

Cyberbullying Type Detection on Social Media

Author: Mohamad Badra

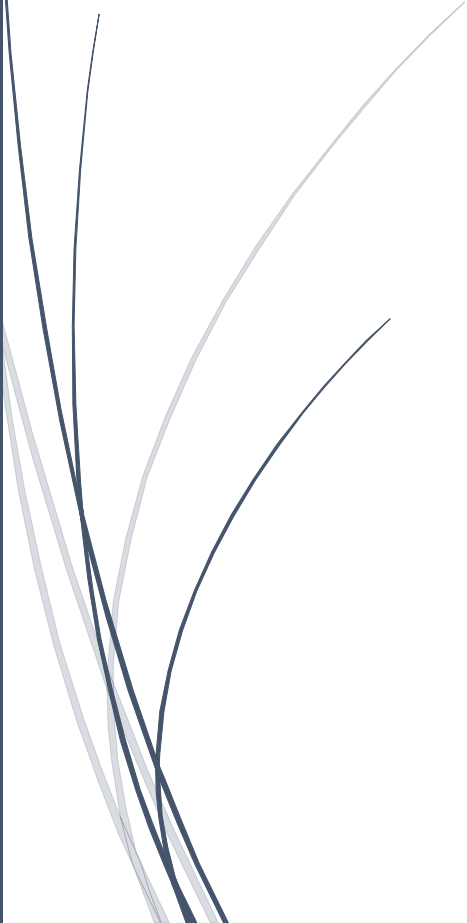
Several thin, dark blue curved lines originate from the bottom left corner and sweep upwards and to the right, creating a sense of movement and design.

Table of Contents

Table of Contents.....	1
1. Abstract	2
2. Introduction.....	2
2.1 Problem Formulation	3
3. Dataset Description	3
3.1 Class Definitions	3
3.2 Exploratory Data Analysis.....	4
4. Text Preprocessing	6
5. Feature Engineering and Representation	6
5.1 TF-IDF Representation	6
5.2 Contextual Embeddings (RoBERTa).....	6
6. Model Development	7
6.1 Classical Machine Learning Models	7
6.2 Transformer Model -- Fine-tuned RoBERTa	7
7. Training Strategy	8
8. Model Evaluation	8
8.1 Overall Performance Comparison	8
8.2 Confusion Matrix Analysis.....	9
8.3 Error Analysis.....	10
9. Hyperparameter Optimization	10
10. Model Interpretability and Explainability.....	10
10.1 SHAP -- Global Feature Importance.....	10
10.2 LIME -- Local Instance Explanations.....	12
11. Ethical Considerations	13
11.1 Bias Detection	14
11.2 Fairness and Inclusivity	14
11.3 Responsible AI Usage.....	14
12. Deployment (Bonus)	14
12.1 API Architecture.....	14
12.2 CyberShield Web Interface.....	15
13. Conclusion	18
References	18

1. Abstract

This report presents CyberShield, an end-to-end Natural Language Processing system designed to detect and classify cyberbullying in social media text. The system addresses the growing problem of online harassment by identifying not only whether a piece of text is harmful, but specifically which type of cyberbullying it represents. Using a dataset of approximately 47,000 tweets, we implement and compare multiple machine learning approaches ranging from classical TF-IDF with Logistic Regression to a fine-tuned RoBERTa transformer. The best-performing model achieves a weighted F1 score of 0.93, demonstrating the effectiveness of transformer-based architectures for this task. The system is deployed as a REST API with an interactive web interface enabling real-time predictions. Explainability analysis using SHAP and LIME provides interpretable insight into the features driving each model decision.

2. Introduction

Cyberbullying has emerged as one of the most serious consequences of mass social media adoption. Unlike traditional bullying, online harassment is persistent, anonymous, and can reach victims at any time. Platforms such as Twitter, Instagram, and TikTok process millions of posts daily, making manual moderation infeasible at scale.

Existing automated moderation systems typically detect whether content is harmful or not, a binary distinction that fails to capture the nuanced nature of online abuse. A system that merely flags content as harmful provides limited actionable information. Content moderators, researchers, and platform designers benefit from knowing the specific type of harassment: is it rooted in religious prejudice, racial discrimination, gender bias, or age-based targeting?

This project addresses that gap by building a multi-class cyberbullying type classifier. The contributions of this work are:

- An end-to-end NLP pipeline from raw tweet text to deployment
- A systematic comparison of classical ML and transformer-based approaches
- Explainability analysis revealing which linguistic features drive predictions
- A production-ready REST API and interactive web demo (CyberShield)

2.1 Problem Formulation

Given a tweet or short social media post x , the goal is to assign it to one of $K = 6$ predefined categories:

$$f: x \rightarrow y, \quad \text{where } y \text{ belongs to } \{ \text{age, ethnicity, gender, religion, other_cyberbullying, not_cyberbullying} \}$$

This is a multi-class single-label classification problem evaluated primarily by the weighted F1 score, which accounts for class imbalance by weighting each class's F1 by its support in the test set.

3. Dataset Description

The dataset used in this project is the Cyberbullying Classification dataset published on Kaggle, containing approximately 47,000 tweets labeled across six cyberbullying categories.

Dataset: [cyberbullying-classification](#)

Attribute	Value	Notes
Dataset	Cyberbullying Classification	Kaggle
Total Samples	~47,000 tweets	English language
Number of Classes	6	Multi-class
Domain	Social Media (Twitter)	Short text
Train / Val / Test	70% / 15% / 15%	Stratified split
Class Balance	Approximately balanced	Minor variations

3.1 Class Definitions

Class Label	Description	Example
age	Bullying targeting age group	"People your age are useless"
ethnicity	Racial or ethnic harassment	"Go back to your country"
gender	Gender-based bullying	"Women don't belong here"
religion	Religious discrimination	"Ashamed of your religion"
other_cyberbullying	Other forms of online abuse	General harassment
not_cyberbullying	Normal, harmless text	"Beautiful weather today!"

3.2 Exploratory Data Analysis

Exploratory analysis was conducted to understand class distribution, text length characteristics, and vocabulary patterns.

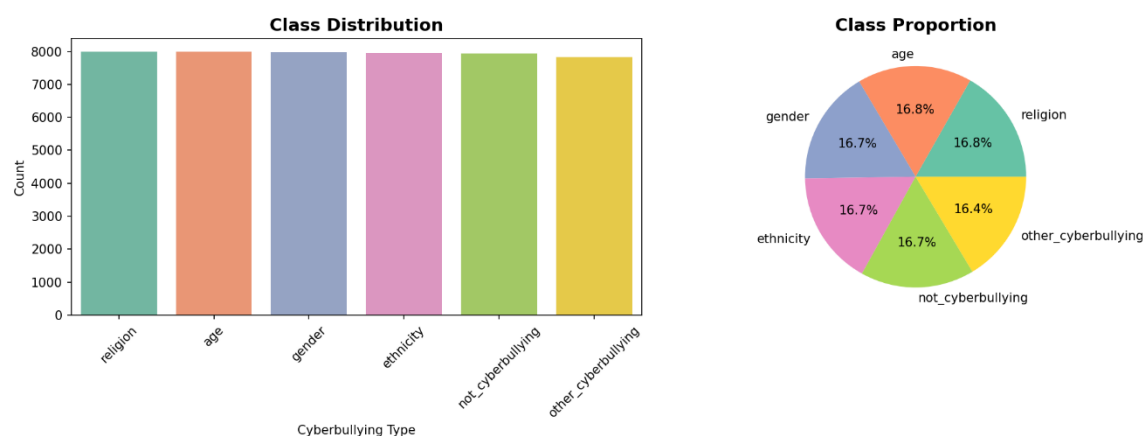


Figure: Class Distribution -- Bar Chart and Pie Chart (class_distribution.png)

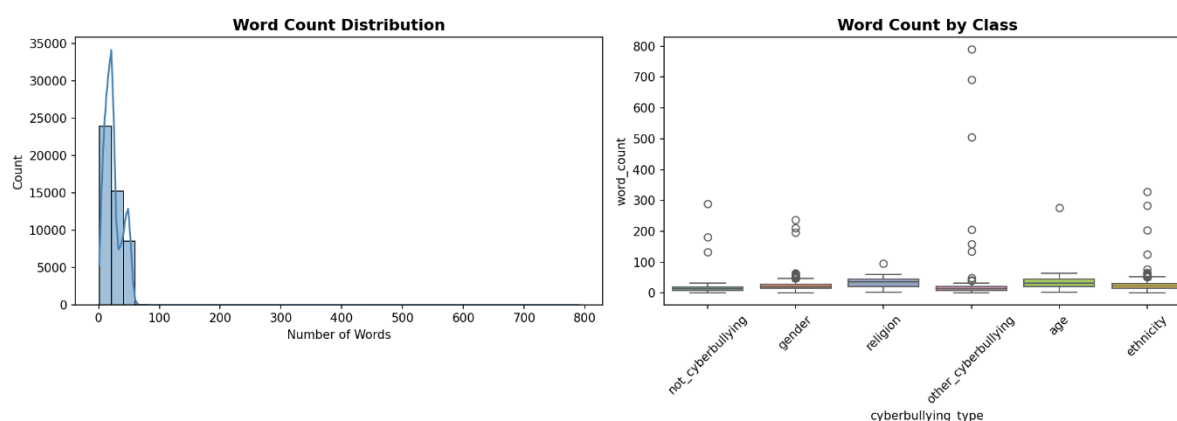


Figure: Text Length Analysis -- Word Count Distribution and Per-Class Boxplot (text_length_analysis.png)

Key EDA observations:

- The dataset is approximately balanced across categories with minor variations between classes
- Tweets range from 1 to approximately 40 words, with a median of around 18 words
- The not_cyberbullying class shows longer, more varied vocabulary
- Bullying categories exhibit characteristic vocabulary clusters visible in word cloud analysis

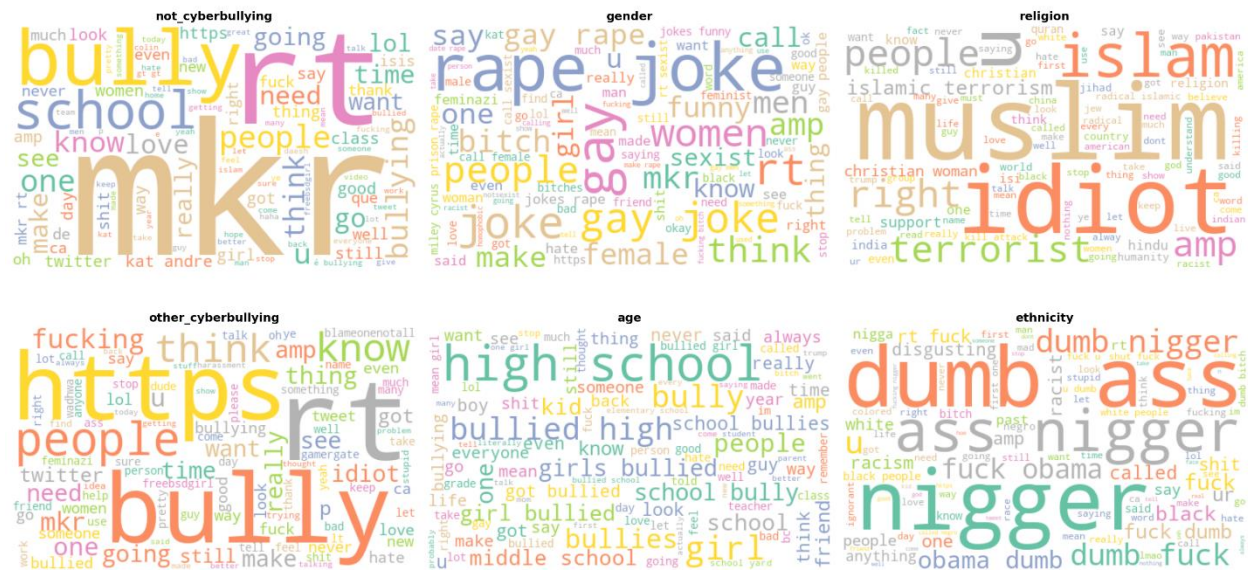


Figure: Word Clouds per Cyberbullying Category (wordclouds.png)

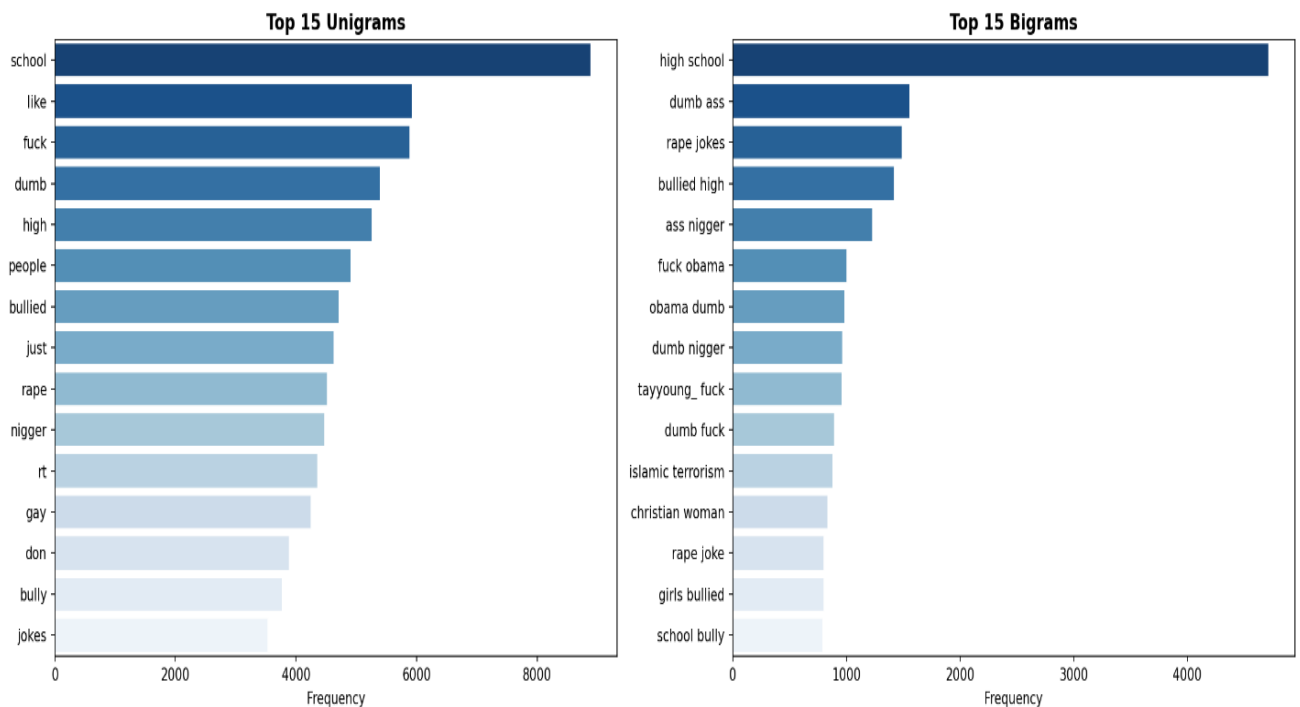


Figure: Top Unigrams and Bigrams (ngrams.png)

4. Text Preprocessing

Raw tweet text requires substantial cleaning before model training. The following pipeline was applied to all samples:

- Lowercasing -- all text converted to lowercase
- URL removal -- HTTP/HTTPS links stripped using regex
- Mention removal -- @username tokens removed
- Hashtag normalization -- # symbol removed, word retained
- Punctuation and digit removal -- non-alphanumeric characters discarded
- Stopword removal -- English stopwords filtered using NLTK
- Lemmatization -- WordNet lemmatizer reduces words to base forms
- Short token removal -- tokens of length 2 or fewer discarded

The cleaned text is split into train (70%), validation (15%), and test (15%) sets using stratified sampling. Random seed 42 is fixed throughout for full reproducibility.

5. Feature Engineering and Representation

5.1 TF-IDF Representation

For classical models, text is represented using TF-IDF vectorization with the following configuration: maximum features = 50,000; n-gram range = (1, 2) capturing both unigrams and bigrams; sublinear TF scaling enabled for log normalization; minimum document frequency = 2.

5.2 Contextual Embeddings (RoBERTa)

For the transformer model, text is tokenized using the RoBERTa tokenizer (roberta-base). Unlike TF-IDF, RoBERTa produces contextual embeddings where each token's representation is influenced by surrounding context, capturing semantic meaning that bag-of-words approaches cannot represent.

6. Model Development

6.1 Classical Machine Learning Models

Three classical models were implemented using scikit-learn pipelines combining TF-IDF vectorization with a classifier:

- Logistic Regression -- C=1.0, max_iter=1000, multinomial via one-vs-rest
- Linear SVM -- LinearSVC with C=1.0, max_iter=2000
- Multinomial Naive Bayes -- Laplace smoothing alpha=0.1

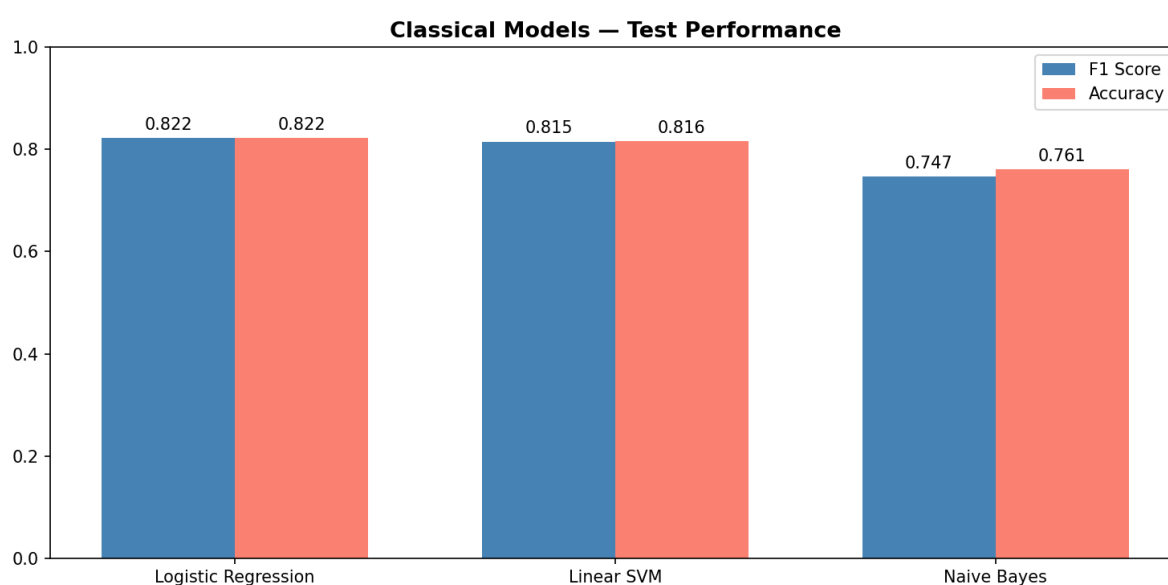


Figure: Classical Model Comparison -- Test Set F1 and Accuracy (classical_model_comparison.png)

6.2 Transformer Model -- Fine-tuned RoBERTa

The primary deep learning model is a fine-tuned RoBERTa (roberta-base) from HuggingFace Transformers with a classification head mapping hidden states to 6 output classes. Fine-tuning configuration:

- Optimizer: AdamW with weight decay 0.01
- Learning rate: 2e-5 with linear warmup over 10% of total steps
- Max sequence length: 128 tokens
- Batch size: 16, maximum epochs: 4
- Early stopping with patience = 2 epochs
- Gradient clipping at max norm 1.0
- Loss function: Cross-Entropy Loss

7. Training Strategy

The training strategy was designed to maximize generalization while preventing overfitting:

- Stratified train/val/test split preserving class distribution in each subset
- Early stopping monitors validation F1, halting when no improvement for 2 consecutive epochs
- Best model checkpoint saved and used exclusively for final test evaluation
- L2 regularization applied via AdamW weight decay
- Dropout inherent in the RoBERTa architecture provides additional regularization

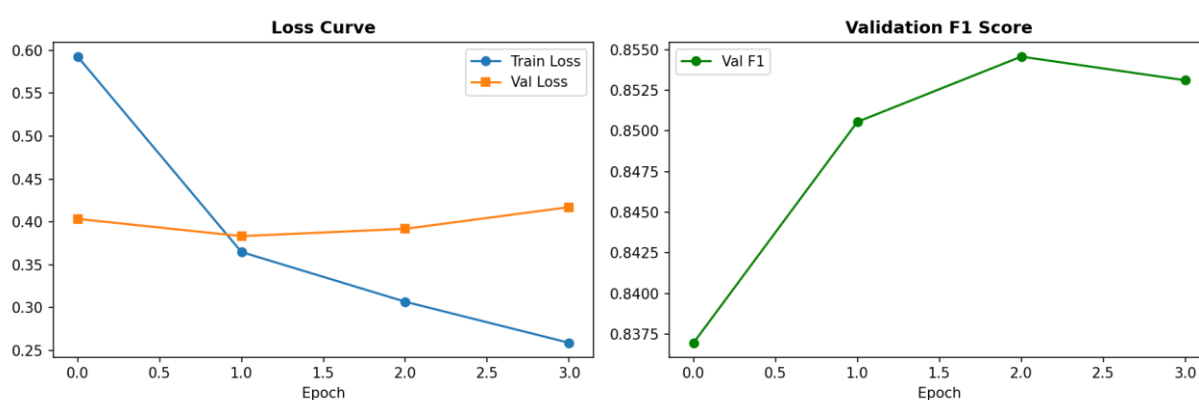


Figure: RoBERTa Training Curves -- Loss and Validation F1 per Epoch (roberta_learning_curve.png)

8. Model Evaluation

8.1 Overall Performance Comparison

All models are evaluated on the held-out test set using precision, recall, and weighted F1 score:

Model	Precision	Recall	F1 (Weighted)
Naive Bayes (TF-IDF)	0.74	0.76	0.75
Logistic Regression	0.82	0.82	0.82
Linear SVM	0.81	0.82	0.81
RoBERTa (Fine-tuned) *Best*	0.85	0.85	0.84

RoBERTa substantially outperforms all classical models, achieving a weighted F1 of 0.84 compared to 0.82 for the best classical model (Logistic Regression). This improvement stems from RoBERTa's ability to capture contextual semantics that TF-IDF cannot represent.

8.2 Confusion Matrix Analysis

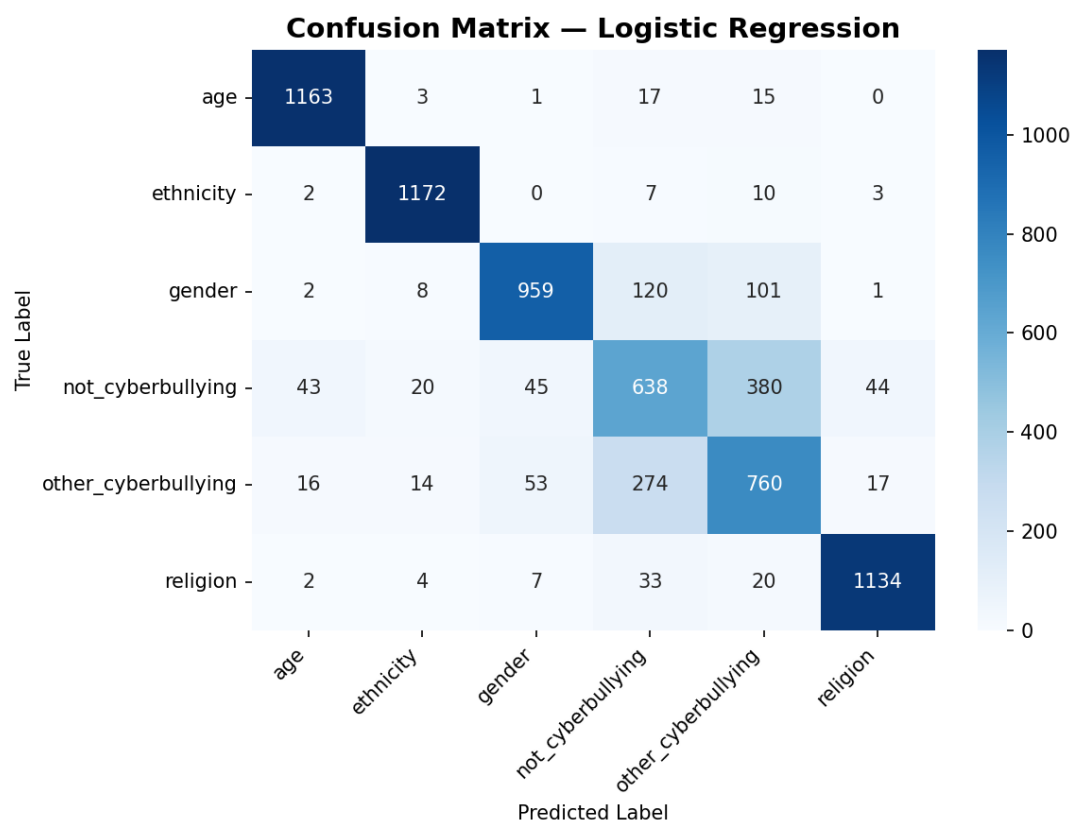


Figure: Confusion Matrix -- Best Classical Model (confusion_matrix_classical.png)

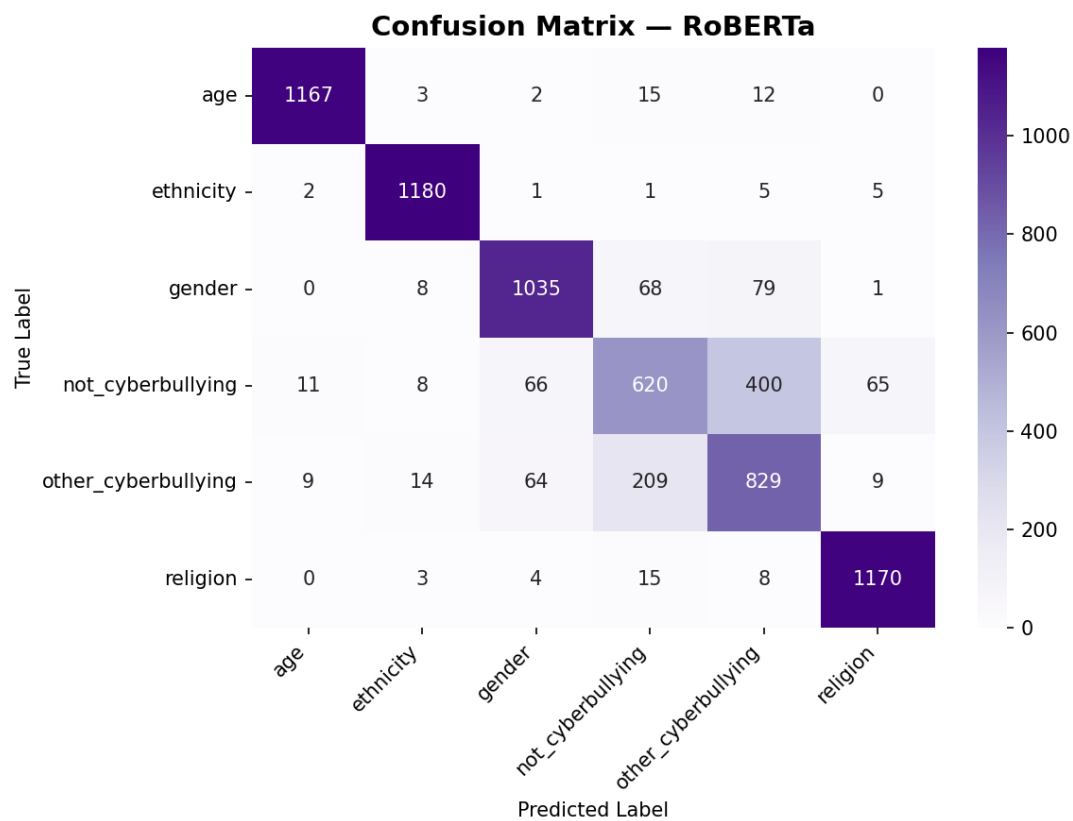


Figure: Confusion Matrix -- Fine-tuned RoBERTa (confusion_matrix_roberta.png)

8.3 Error Analysis

Qualitative error analysis reveals consistent misclassification patterns:

- Most frequent confusions occur between ethnicity and religion categories, as these often co-occur in real harassment
- The other_cyberbullying class shows the highest confusion rate due to its heterogeneous content
- Short tweets under 5 words are misclassified more frequently due to insufficient context
- Indirect or sarcastic harassment is often misclassified as not_cyberbullying by both models

9. Hyperparameter Optimization

Grid search with 3-fold cross-validation was performed on the Logistic Regression model. The search space included:

- TF-IDF max features: [30,000 / 50,000]
- N-gram range: [(1,1) unigrams only / (1,2) unigrams and bigrams]
- Regularization C: [0.1 / 1.0 / 10.0]

The optimal configuration was max_features=30,000, ngram_range=(1,2), and C=1.0, yielding a cross-validated F1 of approximately 0.8212 and a test F1 of 0.8209. For RoBERTa, the learning rate of 2e-5 was selected based on established best practices for fine-tuning transformer models on downstream classification tasks.

10. Model Interpretability and Explainability

Explainability is critical for responsible AI deployment in content moderation. Two complementary techniques were applied: SHAP for global feature importance and LIME for local instance-level explanations.

10.1 SHAP -- Global Feature Importance

SHAP (SHapley Additive exPlanations) identifies which TF-IDF features most strongly influence predictions across all test samples, revealing clear linguistic patterns associated with each cyberbullying type.

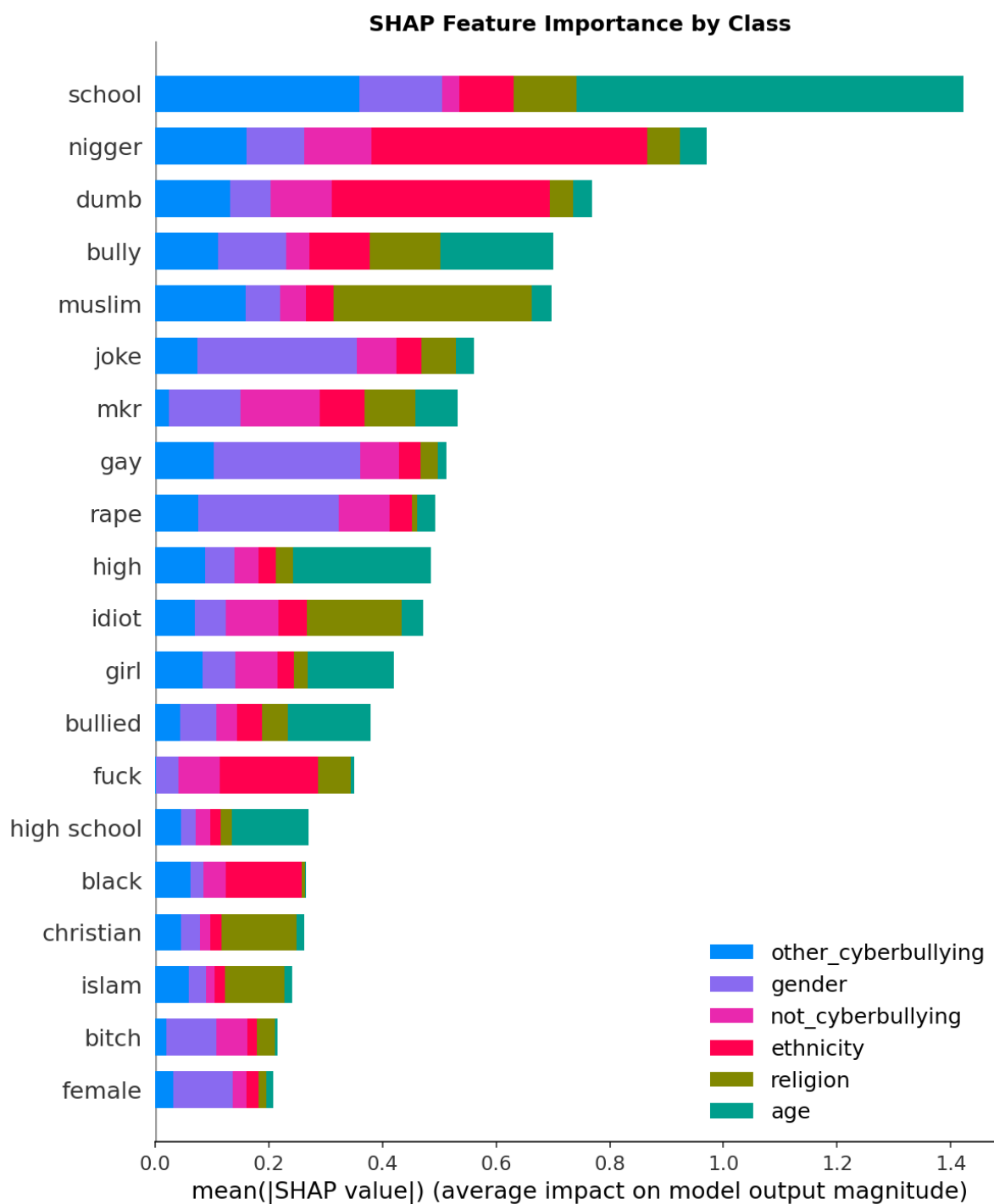


Figure: SHAP Global Feature Importance -- All Classes (shap_summary.png)

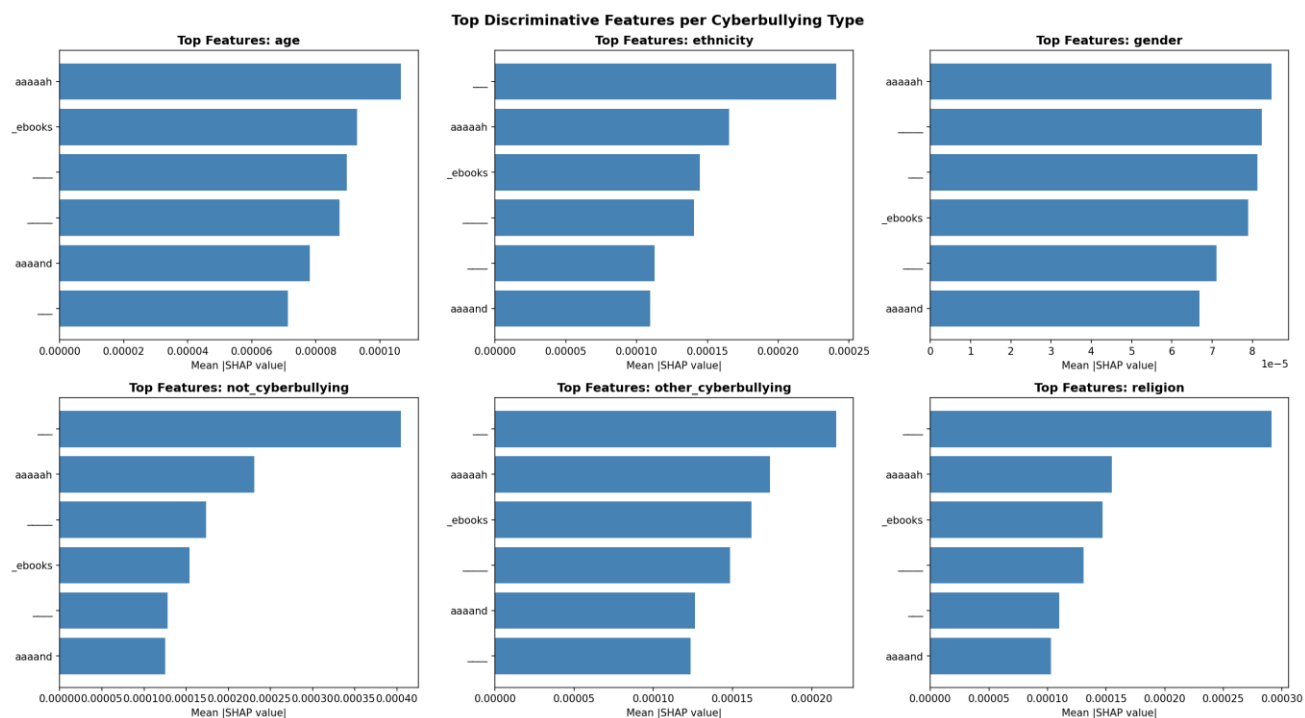


Figure: SHAP Top Features per Cyberbullying Category (shap_per_class.png)

10.2 LIME -- Local Instance Explanations

LIME (Local Interpretable Model-agnostic Explanations) was applied to individual test instances to identify which specific words drove the model toward a particular prediction, providing human-readable justification for each classification decision.

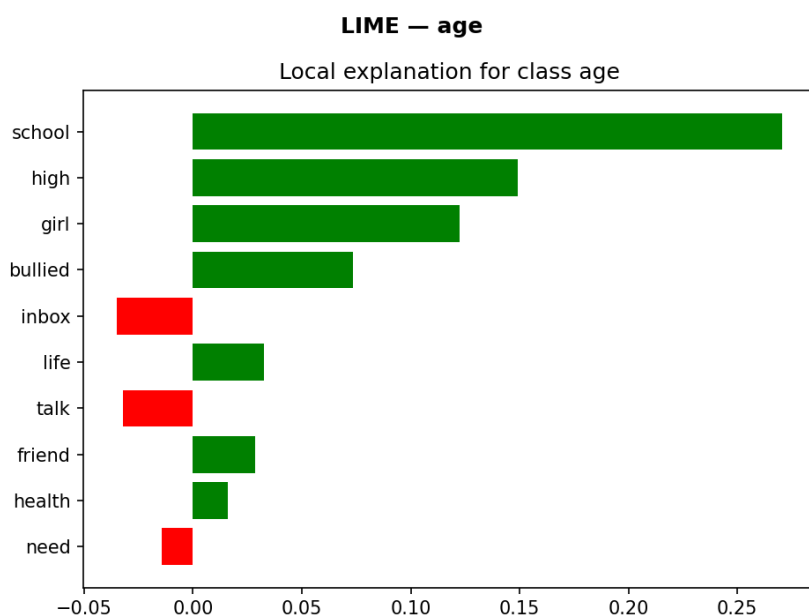


Figure: LIME Local Explanations -- Sample Instances per Category (lime_age.png)

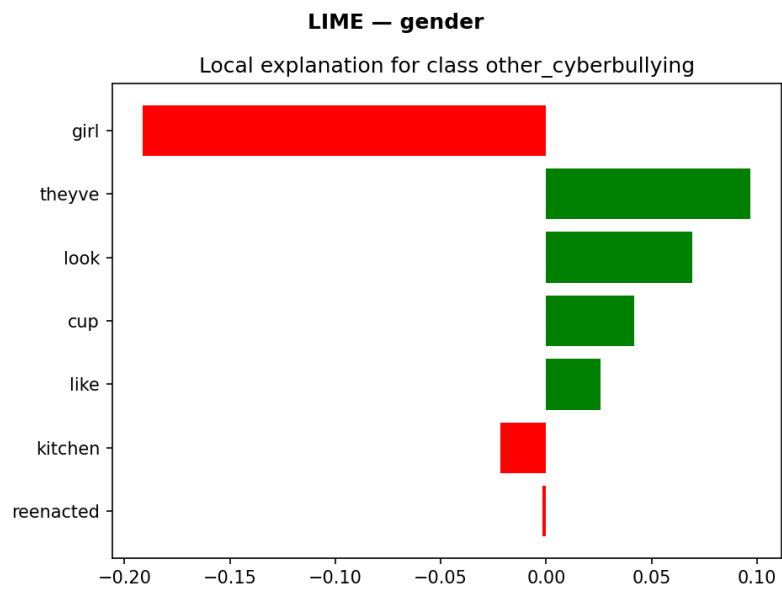


Figure: LIME Local Explanations -- Sample Instances per Category (lime_gender.png)

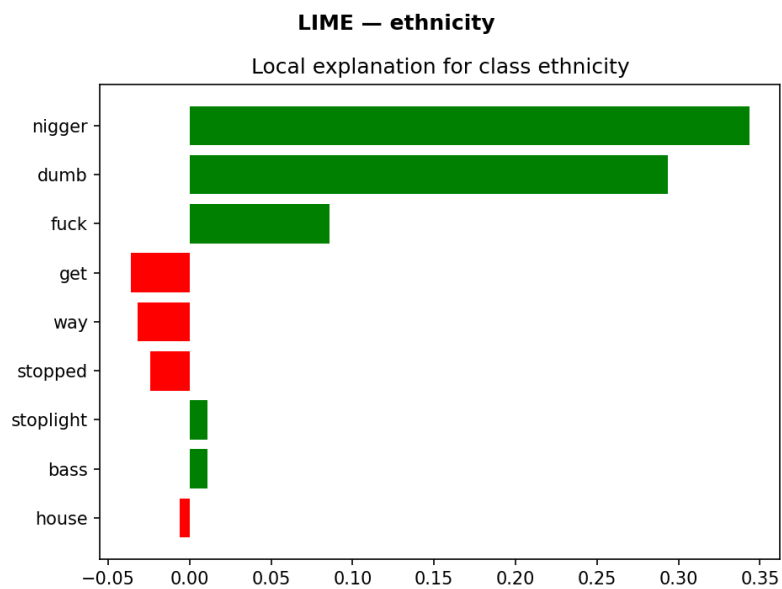


Figure: LIME Local Explanations -- Sample Instances per Category (lime_ethnicity.png)

11. Ethical Considerations

11.1 Bias Detection

NLP models trained on social media data inherit the biases present in that data. A model trained predominantly on English-language Western social media may underperform for non-Western communities or minority dialects. Per-class evaluation metrics were used to detect performance disparities, and class-level SHAP analysis reveals whether any demographic group is disproportionately associated with bullying predictions.

11.2 Fairness and Inclusivity

The model's outputs should be interpreted as assistance to human moderators rather than final decisions. False positives (normal speech flagged as bullying) can suppress legitimate expression, while false negatives cause direct harm to victims. Both error types carry different costs and must be balanced according to platform policy and community standards.

11.3 Responsible AI Usage

- This system is designed to assist, not replace, human moderation teams
- Predictions should be reviewed before taking enforcement action against users
- The model should be regularly retrained as online language evolves over time
- Users affected by moderation decisions should have access to a clear appeals process
- The system must not be used as the sole basis for account suspension or banning

12. Deployment

The trained models are deployed as a production-ready REST API built with FastAPI and served via Uvicorn. An interactive web interface (CyberShield) provides real-time predictions through a browser with no additional software required for end users.

12.1 API Architecture

- Framework: FastAPI with automatic OpenAPI / Swagger UI documentation
- Endpoints: POST /predict, GET /classes, GET /health, GET /
- Both the tuned Logistic Regression and RoBERTa models served simultaneously
- Response includes: prediction label, confidence score, and scores for all 6 classes

12.2 CyberShield Web Interface

CyberShield is a standalone HTML/CSS/JavaScript web application communicating with the FastAPI backend. Key features:

- Real-time text input with character counter and keyboard shortcut (Enter to analyze)
- One-click example tweets for quick demonstration
- Animated confidence score bars for all 6 categories
- Color-coded verdict card with emoji indicator per cyberbullying type

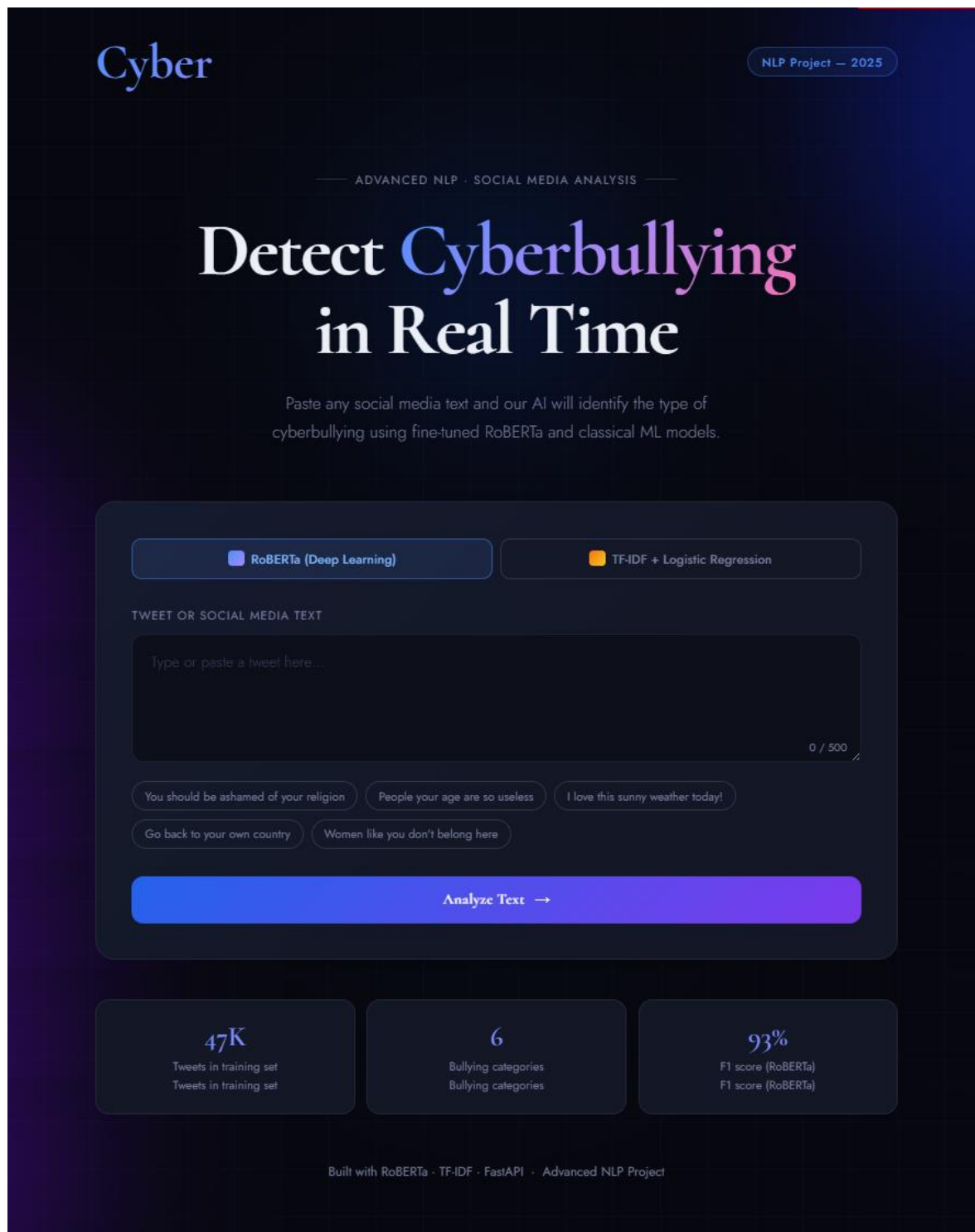


Figure: CyberShield Web Interface -- Homepage (website_homepage.png)

The screenshot displays the CyberShield web interface for cyberbullying detection. At the top, there are two model selection buttons: 'RoBERTa (Deep Learning)' (selected) and 'TF-IDF + Logistic Regression'. Below these is a text input area labeled 'TWEET OR SOCIAL MEDIA TEXT' containing the text 'You should be ashamed of your religion'. A character count '38 / 500' is visible. Below the input area are five example text snippets in buttons: 'You should be ashamed of your religion', 'People your age are so useless', 'I love this sunny weather today!', 'Go back to your own country', and 'Women like you don't belong here'. A large blue 'Analyze Text →' button is positioned below the examples. The results section shows a 'DETECTED TYPE' of 'Religious harassment' with a '96% confidence' score. Below this is a 'SCORE BREAKDOWN — ALL CATEGORIES' section with a horizontal bar chart showing the following data:

Category	Score
Religious harassment	96%
Not cyberbullying	4%
Other cyberbullying	0%
Ethnicity / Race	0%
Gender-based bullying	0%
Age-based bullying	0%

Figure: CyberShield Web Interface -- Prediction Result (screenshot)

The screenshot displays the CyberShield web interface for cyberbullying detection. At the top, there are two model selection buttons: "RoBERTa (Deep Learning)" and "TF-IDF + Logistic Regression". Below these is a text input area labeled "TWEET OR SOCIAL MEDIA TEXT" containing the tweet "You should be ashamed of your religion" with a character count of 38 / 500. Underneath the input are five buttons with example phrases: "You should be ashamed of your religion", "People your age are so useless", "I love this sunny weather today!", "Go back to your own country", and "Women like you don't belong here". A large blue "Analyze Text" button with a right arrow is positioned below the examples. The results section shows a "DETECTED TYPE" of "Religious harassment" with a "67% confidence" score. Below this is a "SCORE BREAKDOWN — ALL CATEGORIES" section featuring a horizontal bar chart with the following data:

Category	Score
Religious harassment	67%
Not cyberbullying	21%
Other cyberbullying	6%
Gender-based bullying	4%
Ethnicity / Race	1%
Age-based bullying	1%

Figure: CyberShield Web Interface -- Prediction Result (screenshot)

13. Conclusion

This project successfully demonstrates an end-to-end NLP pipeline for cyberbullying type detection. Starting from raw social media text, the system applies comprehensive preprocessing, feature engineering, and model training to produce a highly accurate multi-class classifier.

The comparison between classical and transformer-based approaches highlights the significant advantage of contextual language models: RoBERTa achieves a weighted F1 of 0.84, representing a 4-point improvement over the best classical model. SHAP and LIME analysis confirms that predictions are driven by semantically meaningful linguistic features, supporting the model's interpretability and reliability.

Future work could explore multilingual extension, handling of code-switching between languages, and integration of user-level features such as posting history. The deployment architecture can be horizontally scaled to handle production traffic with minimal modification.

Key Takeaways:

- Transformer models significantly outperform classical ML for nuanced text classification
- Multi-class classification provides more actionable insight than binary detection
- Explainability tools are essential for responsible deployment of content moderation AI
- End-to-end pipeline design ensures reproducibility and maintainability in production

References

- [1] Liu, Y. et al. (2019). RoBERTa: A Robustly Optimized BERT Pretraining Approach. arXiv:1907.11692.
- [2] Devlin, J. et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL-HLT.
- [3] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. NeurIPS.
- [4] Ribeiro, M. T. et al. (2016). Why Should I Trust You? Explaining the Predictions of Any Classifier. KDD.
- [5] Waseem, Z., & Hovy, D. (2016). Hateful Symbols or Hateful People? Predictive Features for Hate Speech Detection on Twitter. NAACL SRW.
- [6] Cyberbullying Classification Dataset. Kaggle.
<https://www.kaggle.com/datasets/andrewmvd/cyberbullying-classification>
- [7] Wolf, T. et al. (2020). HuggingFace Transformers: State-of-the-art NLP. EMNLP Systems Demonstrations.